COURSE STRUCTURE FOR B.TECH. Mechanical Engineering FIRST YEAR SEM – I (w. e. f 2024-25)

SEMESTER-I (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-I											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total	
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	Marks	
1	BSC	24MA101T	Mathematics - I	3	1	0	4	4	25	25	50	-	-	100	
2	BSC	24PH102T	Engineering Physics	3	0	0	3	3	25	25	50	-	-	100	
3	BSC	24PH102P	Engineering Physics Laboratory	0	0	2	1	2	-	-	-	25	25	50	
4	BSC	24BT102T	Biology for Engineers	2	0	0	2	2	25	25	50	-	-	100	
5	BSC	24CV101T	Environmental Science	2	0	0	2	2	25	25	50	-	-	100	
6	ESC	24CP101T	Computer Programming - I	1	0	0	1	1	25	25	50	-	-	100	
7	ESC	24CP101P	Computer Programming - I Laboratory	0	0	2	1	2	-	-	-	25	25	50	
8	ESC	24ME101P	Workshop Practices	0	0	2	1	2	-	-	-	25	25	50	
9	ESC	24ME102P	Engineering Graphics	0	0	4	2	4	-	-	-	25	25	50	
10	HSC	24HS101T	English Communication	2	0	0	2	2	25	25	50	-	-	100	
11	HSC	24HS102T	Universal Human Values	1	0	0	1	1	25	25	50	-	-	100	
			Total	**14**	**1**	**10**	**20**	**25**							

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering,
Semester – I

24MA101T					Mathematics – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

Course Objectives

1. To be able to evaluate problems related to differential and integral calculus of complex functions.
2. To be able to obtain area, volume using integral calculus.
3. To be able to formulate and solve various engineering problems using the calculus.
4. To study the properties of sequence and series and to check the convergence and divergence.

Unit – I: Sequence and Series **12 Hrs**

Definition. Convergent, divergent, bounded & monotone sequences. Infinite sums: Basics, Convergence and divergence, Tests for convergence for positive term series, Alternating series – Leibnitz test, Absolute convergence, conditional convergence. Convergence of power series.

Unit – II: Partial Derivatives and its Applications **08 Hrs**

Partial derivative and its application - Euler's theorem - Total derivatives - Jacobians – Maxima and Minima of two variables using Lagrange's multipliers. Taylor's series.

Unit – III: Multiple Integrals and its Applications **12 Hrs**

Definition and evaluation of double integral (Cartesian – Polar form) – Change of orders - Change of variables – Evaluation of triple integral, change of variables (Cartesian to spherical – and cylindrical) – Applications, area – volume – center of mass – center of gravity by double and triple integral.

Unit – IV: Vector Calculus **12 Hrs**

Gradient, divergence and curl – Directional derivative – Irrotational and Solenoidal vector fields – Vector Integration – Simple problems on line, surface and volume integrals – Green's theorem in a plane, Gauss divergence theorem and Stokes' theorem (without proofs) – Simple application involving cubes and rectangular parallelepipeds.

Total **42 Hrs**

Course Outcomes

On completion of the course, student will be able to:

CO1: Identify the use of convergence of infinite series in engineering aspects.
CO2: Understand the concept of Directional derivative, Irrotational and Solenoidal vector fields.
CO3: Apply the concept of differential and integral calculus in engineering problems.
CO4: Analyze the obtained solution in linear and non-linear domains.
CO5: Appraise mathematical problems from complex domain.
CO6: Evaluate problems on Green's, Stokes' and Divergence theorems.

Text/Reference Books

1. B. S Grewal, "Higher Engineering Mathematics", Khanna Pub., Delhi.
2. R. K. Jain and S. R. K. Iyengar, "Advanced Engineering Mathematics", Alpha Science.
3. Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley.
4. G. Strang, "Linear Algebra and its applications", Cengage Learning.
5. K. Hoffman and R. A. Kunze, "Linear Algebra", Prentice Hall of India.
6. B.V. Ramana, "Higher Engineering Mathematics", McGraw Hill.

Pandit Deendayal Energy University, School of Technology
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Semester – I

24PH102T					Engineering Physics (For ME, Civil)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

Course Objectives

1. To master the concepts of vector algebra and the fundamentals of mechanics.
2. To explore the principles of elasticity in solids, band theory of solids and the Physics of semiconductors.
3. To investigate the behaviour of waves and oscillations, along with their applications in acoustics and ultrasonics.
4. To gain insights into thermal physics, covering conduction, convection, and radiation, and their practical applications.

Unit – I: Mechanics **12 Hrs**

Introduction to vector algebra, Vector fields and scalar fields, Concepts of gradient, Divergence and Curl, Gauss-Divergence Theorem, Stokes' theorem and its applications, The everyday forces of Physics, Work-energy theorem, Rigid body motion and the conservation of angular momentum, Center of mass, Center of gravity, Moment of inertia: Definition, Moment of inertia of simple planar laminas.

Unit – II: Physics of Solids **10 Hrs**

Elasticity, Stress & Strain, Hook's law, Young's modulus, Poisson's ratio, Rigidity modulus, Relationship between elastic constants, Properties of solids, electrical conductivity, Wiede-mann Franz Law, Band structure of conductor, insulator and semiconductor, Intrinsic and extrinsic semiconductor, Conductivity of semiconductors, charge concentrations, Hall effect, LED and Solar cells, Magnetism and its origin, Magnetic properties of solids, Physics of superconductors, Applications.

Unit – III: Waves, Oscillations and Acoustics **08 Hrs**

Waves and oscillations: Types of waves, Simple harmonic motion, Damped simple harmonic motion, types of damping, Forced oscillation, Resonance, Energy Transport in Wave motion. Acoustics & Ultrasonic: Introduction to Sound, Sabine's reverberation theory, Acoustical defects and their remedies, Doppler Effect, Ultrasonic waves, Methods of their generation and detection, Properties and applications of ultrasonic waves.

Unit – IV: Thermal Physics **12 Hrs**

Thermodynamic systems (closed and open), Thermodynamic properties and equilibrium, Concepts of heat and work, Laws of thermodynamics, Carnot cycle, Heat transfer-thermal expansion of solids and liquids, Conduction in solids, Thermal conductivity, Forbe's method, Lees' disc method, Thermal insulation and its applications, Thermal Convection and its applications, Newton's law of cooling, Thermal Radiation – emission and absorption radiation, emissive power, Black body radiation, Stefan's laws, Wien's law.

Total **42 Hrs**

Course Outcomes

On completion of the course, student will be able to:

CO1: Describe fundamental concepts of mechanics, such as vector algebra, everyday forces of Physics, Moment of inertia, etc.
CO2: Explain the principles underlying the physics of solids, including elasticity and the behavior of energy bands.
CO3: Solve problems involving mechanics, solids, waves, oscillations, and acoustics.
CO4: Analyze the behavior of waves, oscillations, and thermal phenomena, including their practical applications.
CO5: Critically evaluate the significance of thermodynamic laws and superconductivity theories in solving real-world problems.
CO6: Develop solutions and innovative applications in mechanics, solids, waves, and thermal physics by integrating concepts and principles learned throughout the course.

Text/Reference Books

1. D. Kleppner and R. J. Kolenkow, An Introduction to Mechanics, Tata McGraw-Hill.
2. D. Halliday, R. Resnick, J. Walker, Fundamentals of Physics, John Wiley & Sons.
3. Sears and Zemansky, University Physics, Pearson publications.
4. M. N. Avadhanulu, P.G. Kshirsagar, A textbook of Engineering Physics, S Chand & Co. Ltd.
5. F. M. A. Wahab, Solid State Physics: Structure and Properties of Materials, Narosa Publishing House Pvt. Ltd.
6. Richard, Oscillations and Waves: An Introduction, Taylor & Francis.
7. Y. A. Cengel, Introduction to Thermodynamics and Heat Transfer, Tata McGraw-Hill.
8. B. Lal and N. Subramaniyam, Heat and Thermodynamics, S Chand & Co. Ltd.

Pandit Deendayal Energy University, School of Technology
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Semester – I

24PH102P					Engineering Physics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

Course Objectives

1. To comprehend the operation of diverse electrical, mechanical, and optical instruments utilized within the laboratory setting.
2. To grasp fundamental principles of Physics and apply them effectively in the execution of experiments.
3. To acquire hands-on experience in Physics through experimental work, fostering practical understanding.

List of Experiments

1. To determine the acceleration due to gravity 'g' using a simple pendulum.
2. To determine the acceleration due to gravity 'g' using a bar pendulum.
3. To determine the electrical conductivity of metals.
4. To study the characteristics of Si solar cells.
5. To explore the usage of Digital Oscilloscope.
6. To study the phenomenon of photoconductivity using CdS photo-resistor.
7. To determine linear thermal expansion coefficient of solid bodies.
8. To study the phenomenon of Hall Effect.
9. To determine volumetric coefficient of expansion of liquids.
10. To study the reflection of ultrasonic waves.
11. To demonstrate/investigate resonance in forced oscillations.
12. To study the principle of heat pump.
13. To determine energy band gap of semiconductor using four probe method.
14. To measure the slit width of single, blade and double slits.
15. To determine e/m using Thomson's method.

** Any 10 experiments will be conducted.

Course Outcomes

On completion of the course, student will be able to:

CO1: Apply mathematical reasoning to analyze the concepts of mechanics.
CO2: Compare and contrast experimental results to determine electrical and thermal conductivity.
CO3: Demonstrate and implement the concepts of waves and oscillations.
CO4: Investigate the electrical properties of a given semiconductor device.
CO5: Examine the charge transport mechanism in different conductors.
CO6: Analyze the behavior of waves, oscillations, and thermal phenomena in experiments.

Text/Reference Books

1. D. Halliday, R. Resnick, J. Walker, Fundamentals of Physics, John Wiley & Sons.
2. C. Kittel, Introduction to Solid State Physics, John Wiley.
3. C.S. Robinson, R, Das, Textbook of Engineering Physics Practical, University Science Press.
4. Kittel, Knight and Ruderman, Mechanics - Berkeley Physics Course, Tata McGraw-Hill.
5. Avadhanulu, A text book of engineering Physics, S. Chand & Company, Ltd.
6. Brij Lal, N. Subrahmanyam, Heat and Thermodynamics, S. Chand & Company, Ltd

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering,
Semester – I

24BT102T					Biology For Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

Course Objectives

1. To understand the basic biological concepts and their engineering applications.
2. To introduce the students with an understanding of biodesign principles to create novel devices and structures.
3. To provide the students an appreciation of how biological systems can be re-designed as substitute products for natural systems.
4. To study the development of interdisciplinary vision of biological engineering

Unit I: Introduction To Biomolecules — 7 Hrs.

Cell as a basic unit of life, prokaryotic and eukaryotic cells, microbes, plant and animal cells; Cell organelles – structure and function; Cell membrane. Vitamins and Minerals, Carbohydrates, Lipids, Amino Acids and proteins, Nucleic Acids (DNA and RNA).

Unit II: Nature-Bioinspired Mechanisms — 7 Hrs.

Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics). The eye functions as a camera. Heart as a pump system, The ear and nose as signal transmission models. Lungs operate as an exchange model, kidneys function as a filtration system. Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro).

Unit III: Trends In Bioengineering — 7 Hrs.

Stem cell and its applications, Bio printing techniques and materials. Applications of Bioinformatics, Artificial Intelligence for diagnosis and detection of communicable and non-communicable diseases, Biosensors in healthcare, Environmental monitoring, Food Safety and Biosecurity.

Unit IV: Engineering Perspectives Of Biological Sciences — 7 Hrs.

Biology and engineering crosstalk – At cell level: Hybridoma technology, At tissue level: Plant Tissue Culture, Animal Tissue Culture; Tissue Engineering, Introduction to Biomimetics and Biomimicry, Nanobiotechnology, Introduction to Radiology, High-throughput diagnostics in clinics: Molecular Diagnostics (PCR), DNA chips.

Total 28 Hrs

Course Outcomes

On completion of the course, student will be able to:

- **CO1:** Gain fundamental knowledge of origin of life.
- **CO2:** Demonstrate the involvement of major organs and systems in the human body as models for bioengineering design.
- **CO3:** Analyse the scope and opportunities of Implementing bioengineering concepts relates to advanced imaging and diagnostic techniques for clinical diagnosis and disease management.
- **CO4:** Get acquainted with concepts of bioengineering trends including biosensors, AI , Imaging techniques in disease diagnosis and treatment
- **CO5:** Focus on implications of Bioinspired designs, Bioengineering trends including, tissue engineering and molecular diagnostics.
- **CO6:** Think critically towards exploring innovative biobased solutions for socially relevant problems.

Text/Reference Books

1. Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jaganthan M., Biology for Engineers, Tata McGraw-Hill, New Delhi.
2. Arthur T. Johnson, Biology for Engineers, CRC Press, Taylor and Francis.
3. Geoffrey M.Cooper, The Cell: A molecular Approach, ASM Press.
4. Sohini Singh and Tanu Allen, Biology for Engineers", Vayu Education of India, New Delhi.
5. Yoseph Bar-Cohen, "Biomimetics: Nature-Based Innovation, CRC Press.
6. Stuart Fox, Krista Rompolski, Human Physiology", McGraw-Hill (eBook).
7. D. Floreano and C. Mattiussi, Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies, MIT Press.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering,
Semester – I

24CV101T					Environmental Science					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

Course Objectives

1. To develop a comprehensive perspective of environment and sustainable development
2. To understand the causes and effects of various types of pollution
3. To develop an understanding of the various strategies for controlling the pollution
4. To introduce the emerging environmental domains

Unit I : Introduction To Environment — 05 Hrs.

Sustainable Development; Sustainable Development Goals; Environmental Studies – Its importance and Multidisciplinary nature, Introduction to Environmental Parameters and their standards (air, water, soil, noise, etc.); Ecosystem and its types, Ideal ecosystem, Biodiversity : Its importance and conservation.

Unit II : Multi-Scale Environmental Pollution (Global, Regional And Local) — 06 Hrs.

Pollution, Causes and Effects of different types of pollution : Air Pollution, Water Pollution, Soil Pollution, Solid Waste (organic and Inorganic) Pollution, Hazardous Waste Pollution, Marine Pollution, Noise Pollution, Thermal Pollution, Radioactive Pollution; Introduction to man-made disasters like floods, heat waves, landslides, etc., Introduction to the various instruments for measuring air pollution, water pollution, noise, etc.

Unit III : Environmental Pollution Control Strategies — 09 Hrs.

Multi-approaches for reducing various types of pollution: Introduction to Water and Wastewater treatment technologies, Air and Noise pollution control techniques, Introduction to different environmental management concepts like Swachh Bharat Mission, Mission LiFE (Lifestyle For Environment), etc. Indian Culture and Traditional Wisdom for managing environment

Unit IV: Emerging Environmental Managements Domains — 08 Hrs.

Concept of Zero Liquid Discharge (ZLD) and the reuse of the treated wastewater, Green Credit Rules - 2023, Clean Development Mechanisms (CDM) and Carbon Credits, Green Buildings, Carbon Footprint and Water Footprint, Green Business, International Environmental Laws, Environmental Auditing.

Total 28 Hrs

Course Outcomes

On completion of the course, student will be able to:

- **CO1:** Demonstrate comprehension of sustainable development and environmental aspects.
- **CO2:** Recognize the interdisciplinary characteristics inherent in Environmental studies.
- **CO3:** Evaluate the impact of various pollutants on the environment.
- **CO4:** Assess the efficacy of different technologies for environmental pollution control.
- **CO5:** Analyze different environmental management policies and their implications.
- **CO6:** Synthesize knowledge about emerging environmental management paradigms.

Text/Reference Books

1. Bharucha Erach, Textbook for Environmental Studies, UGC New Delhi.
2. Daniel B. Botkin & Edwards A. Keller, Environmental Science, Wiley India edition.
3. Miller T. G. Jr., 2006. Environmental Science, Clengage Learning.
4. R. Rajagopalan, Environmental Studies, Oxford University Press.
5. Gilbert Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, PHI.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering,
Semester – I

24CP101T					Computer Programming – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

Course Objectives

1. To understand the usage of operators and data types.
2. To apply different types of Conditional and looping statement.
3. To create different types of data collections.
4. To implement user defined function.
5. To perform different operations upon files.

Unit – I: Basics of Programming **03 Hrs**

Introduction to Computer Programming, Features of C language, Structure of C program, program execution flow, C Tokens, variables, Data types, Operators, Decision control statements-if, switch, go to statement. Loop control structures- while, do-while, for loop, Break statement, Continue statement.

Unit – II: Derived Data types **04 Hrs**

Array: one dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, String-Basic Concepts, Inbuilt String manipulation Functions, Pointer, Pointer arithmetic, Pointer to pointer, Array of Pointers

Unit – III: Functions and Structures **04 Hrs**

Introduction to user defined functions, Types of Functions, Call by value-call by reference, recursion, pointers to functions, Structures, Array of Structure, Union.

Unit – IV: Files Handling **03 Hrs**

File handling in C, Different types of files, Operations on Files such as File creation, File deletion, File access modes such as read, write, append, File concatenation, File handling using seek function..

Total **14 Hrs**

Course Outcomes

On completion of the course, student will be able to:

- **CO1:** Understand basics of programming.
- **CO2:** Identify different programming constructs for a problem.
- **CO3:** Apply appropriate derived data type for data storage.
- **CO4:** Prepare a user defined data type based on data attributes.
- **CO5:** Construct user defined functions for problem solving.
- **CO6:** Analyse different data structure based on application requirement.

Text/Reference Books

1. Kernighan & Ritchie, C Programming Language, PHI
2. K. N. King , C Programming: A Modern Approach, W.W. Nortorn
3. David Griffiths and Dawn Griffiths, Head First C: A Brain-Friendly Guide, O'Reilly
4. E.Balaguruswamy, Programming in ANSI C, McGraw-Hill
5. Y.P. Kanetkar, Let us C, BPB Publication
6. Y.P. Kanetkar , Pointers in C, BPB Publications

Pandit Deendayal Energy University, School of Technology
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Semester – I

24CP101P					Computer Programming Laboratory – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

Course Objectives

1. To implement basic programming concepts.
2. To create different types of data collections.
3. To implement user defined function.
4. To perform different file handling operations.

List of Experiments:

1. **Introduction to Computer Programming**: Understanding compilation process through a simple C program, program execution flow, C Tokens, variables and keywords and identifiers, types of C constants and variables.
2. **Simple and formatted Input Output Operations:** Data types, Operators, Input /output statements in C, Formatted I/O, format specifiers, escaper sequences
3. **Decision making and branching:** if, if-else, if-else ladder, switch, go to statement, conditional operator statement
4. **Looping control structures:** while, do-while, for loop, Break statement, Continue statement
5. **Derived Data Type: Array and Strings:** One dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, Basic Concepts, Inbuilt String manipulation with and without using inbuilt functions.
6. **Derived Data Type: Structure and Union:** structure, arrays and structures, structures and functions, pointer to structure, typedef, unions
7. **Functions:** Introduction to user defined functions, Types of Functions, Call by value-call by reference, header file creation, recursion, pointers to functions, arrays and functions
8. **Pointers:** Pointer's basics, use of &, * operator in context to pointers, Pointer arithmetic, Array and String processing using pointer, pointer to pointer, Array of Pointers
9. **File Handling in C:** File handling in C, Different types of files, Operations on Files, File handling functions.

Course Outcomes

On completion of the course, student will be able to:

CO1: Identify the use of appropriate naming conventions and programming style including appropriate comment density.
CO2: Implement a basic C program using appropriate control structure.
CO3: Apply appropriate derived data types based on data attributes.
CO4: Develop solutions with pointers and utilize them to access strings and structures.
CO5: Design user defined functions for problem solving and reuse them across different programs.
CO6: Apply suitable file handling functions and operations.

Text/Reference Books

1. Kernighan & Ritchie, C Programming Language, PHI.
2. K. N. King , C Programming: A Modern Approach, W.W. Norton.
3. E. Balaguruswamy, Programming in ANSI C, McGraw-Hill.
4. Y.P. Kanetkar,Let us C, BPB Publication.
5. Y.P. Kanetkar, Pointers in C, BPB Publication.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering,
Semester – I

24ME101T					Workshop Practice					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

Course Objectives:

1. To understand safety in various manufacturing processes.
2. Learn how to use various measuring tools for engineering applications.
3. Hands on training and preparation of job using wood, metal and sheet as per drawing.
4. Understand various manufacturing processes like machining, welding, soldering and 3D printing for prototypes.

List of Experiments

1 Introduction to Workshop safety, layout and identification of various materials- plastic, wood, metals-ferrous and nonferrous, rubber, glass etc.
2 Use of measuring tools for engineering applications
3 Fitting job: Detailed drawing of work piece, use of fitting tools and job preparation.
4 Carpentry job: Detailed drawing of work piece, use of carpentry tools and job preparation.
5 Sheet metal job: Detailed drawing of work piece, use of sheet metal working tools and job preparation.
6 Plumbing job: Internal/External threading, piping network using Tees, Elbows, Reducer, Bends etc
7 Hands on training on mini lathe and milling machine
8 Demonstration of welding, brazing and soldering
9 Soldering and desoldering for PCB
10 3D printing using polymer and metal.

Course Outcomes

On completion of the course, student will be able to:

CO1 : Define workshop safety and various engineering materials
CO2 : Understand various measuring equipment
CO3 : Apply various workshop tools in preparing job for carpentry, fitting, sheet metal and plumbing
CO4 : Examine various manufacturing operations like welding and machining
CO5 : Evaluate soldering operation for PCB
CO6 : Create prototype using 3D printing

Text/Reference Books

1. S. K. Hajra Choudhury, Elements of Workshop Technology, Vol. I & II, Media Promoters and Publishers.
2. H. S. Bawa, Workshop Practice, Tata-McGraw Hill.
3. Kalpakjian S. And Steven S. Schmid, Manufacturing Engineering and Technology, Pearson Education India Edition.
4. Rao P.N., Manufacturing Technology, Vol. I and Vol. II, Tata McGraw Hill House.

24ME102T					Engineering Graphics - Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	4	2	4	--	--	--	50	50	100

Course Objectives

1. To cover the fundamental of engineering drawing and standards used in drawing.
2. To explain the students to communicate ideas using orthographic and isometric projection methods.
3. To help students to use CAD software to prepare drawings.
4. To demonstrate the presentation of drawing using sketching and 3D modelling in CAD tool.

List of Experiments

1 Introduction to Engineering Graphics: Principles of engineering graphics and their significance, drawing instruments & accessories, lettering and numbering, types of lines, dimensioning methods, basic geometric drawing, reading a drawing.

2 Orthographic Projection: Introduction to projection, types of projection, 1st angle and 3rd angle projection

3 Isometric Projection: Principles of isometric projection – isometric scale, isometric views, conventions, conversion of isometric views to orthographic views and vice-versa

4 Projection of Solids and Development of Surface: Classification of solids, projections of solids like cylinder, cone, pyramid, and prism with its inclination to reference plane, development of surfaces of right regular solids - prism, pyramid, cylinder and cone.

5 Introduction of Computer Aided Engineering Drawing: Demonstrating knowledge of the theory of CAD software, use of software in drawing, CAD software user interface, commands, Coordinate System, menus and toolbars, planes, dimensioning, saving of files, Select and erase objects, zoom tools, and others

6 Basic sketching using CAD tool: Sketch entities using tools – origin, points, lines, circle, arcs, polygons, fillets and chamfer, trim, extend and offset, copy, cut, delete and others

7 Advanced sketching using CAD tool: Sketching entities using relation constrains, Mirror, Patterning, full definition of drawing and others

8 Basic 3D modelling using CAD tool: Extrude, cut, drawing on different planes, editing, symmetric, revolving, and others

9 Computer aided drawing sheets: Preparing drawing sheets, creating different views, section view, drawing templates, and others

Course Outcomes

On completion of the course, student will be able to:

CO1 : Recalling the fundamentals of engineering graphics by considering basic rules of drawing, dimensioning, and labelling.
CO2 : Explain the principle of projection using orthographic and isometric projection.
CO3 : Represent the 2-dimensional drawing using CAD tool.
CO4 : Construct the 3-dimensional geometries using CAD tool.
CO5 : Apply the concept of engineering drawing by organizing drawing views and applying necessary dimensions by preparing drawing sheets
CO6 : Analyse the intricate details of solid using projection of solid, sectioning of solid and development of lateral surfaces.

Text/Reference Books

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing.
2. Shah P.J., Engineering Graphics, S. Chand Publishing.
3. Agrawal, B. & Agrawal C. M., Engineering Drawing, Tata McGraw Hill Publishers.
4. Hanifan R, Perfecting Engineering and Technical Drawing, Springer International Publishing Switzerland.
5. Corresponding Set of CAD Software Theory and User Manuals.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering,
Semester – I

24HS102T					Universal Human Values					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	--	--	100

Course Objectives

1. To understand the need of nurturing human values through the process in value-based education system.
2. To understand and develop a holistic perspective on self-exploration and being in harmony with family, society and nature.
3. To facilitate the students in understanding harmony at all the levels and applying in their profession and work place to lead an ethical life.

Unit I: Human Values And Process Of Value Education **04 Hrs.**

Human values, human aspirations and the ultimate goal, understanding happiness and prosperity, appraise the meaning of satisfaction and happiness in current scenario, harmony and compatibility, values imbibed education system and process

Unit II: Knowing Self - Harmony With Self **04 Hrs.**

Understanding self, capabilities and challenges, understanding material (physical facilities) and spiritual needs - need of mind and body, understanding body as an instrument, harmony between mind and body, synchronizing physical health and mental health, practicing healthy habits for healthier me

Unit III: Harmony In Relationship – Family, Society And Nature **03 Hrs.**

Harmony in relationships, values for harmony in any human-human interaction, harmony in family, and society, trust and respect for others, self esteem and ego, equality, equity, inclusion and liberation, concept of '*Vasudhaiva Kutumbakam',* understanding co-existence and sync with nature

Unit IV: Harmony In Profession And Ethical Behavior **03 Hrs.**

Ethical human conduct, acceptance and respect, appraising the qualities of others, professional competence for enabling harmony in system and enabling universal human order, scope of eco-friendly systems, strategies to reach the harmonious ecosystem to reach Universal Human Order 'Sarvabhauma Vyavastha'

Total **14 Hrs**

Course Outcomes

On completion of the course, student will be able to:

CO1: Understand the significance of human values, its need, and process of value education.
CO2: Appraise the meaning of happiness and prosperity as short- and long-term goal of life. Understand them and in context of the current scenario
CO3: Distinguish between the mind and body, physical and spiritual wellbeing for harmony within self
CO4: Assess the value of harmonious relationship based on trust, respect and enduring its role in all human-human relationships to build harmonious society
CO5: Understand the importance of harmony with nature and appreciate co-existence for harmonious ecosystem.
CO6: Create the perfect professional place and work environment following the ethical practices and strategize to uphold the human values at all the levels and interactions.

Text/Reference Books

1. R. R. Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and Professional Ethics, Excel books.
2. A. Nagraj, Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
3. A. N. Tripathi, Human Values, New Age Intl. Publishers.
4. M. K. Gandhi. The Story of My Experiments with Truth, Fingerprint Publishing.
5. Ivan Illich, Energy & Equity, The Trinity Press, Worcester, and Harper Collins.
6. E. F. Schumacher, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain.
7. Sussan George, How the Other Half Dies, Penguin Press.